Student Performance & Learning Analytics

Project Report | Data Science Workflow, EDA, Feature Engineering & Predictive Modeling

Project purpose: Analyze student learning and academic indicators to explore performance patterns and demonstrate an introductory prediction workflow.

Dataset notice: This deliverable uses a reproducible synthetic dataset created for practice. It does not represent real students or any actual institution. Subject-wise marks were not supplied in the brief, so subject-level analysis is documented as an extension rather than fabricated.

1. Objectives

- Inspect data structure, data types, missing values, duplicate records, and descriptive statistics.
- Clean data by removing duplicates, imputing missing values, and validating percentage ranges.
- Explore score distribution and relationships between attendance, study time, prior scores, assignments, and final score.
- Create visualizations and attendance/study categories for interpretation.
- Train a Linear Regression model for score prediction and a Logistic Regression model for illustrative pass/fail classification.
- Evaluate models and document limitations, reproducibility, and ethical considerations.

2. Tools and Technologies

Python, Pandas, NumPy, Matplotlib, scikit-learn, Jupyter Notebook, CSV/Excel-compatible files, and Git/GitHub. Seaborn, Tableau, or Power BI can be added for further visualization and dashboard work.

3. Dataset Description

Column	Meaning
student_id	Synthetic student identifier
study_hours_per_day	Approximate daily study hours
attendance_percent	Attendance percentage
previous_score	Previous academic score
assignment_completion_percent	Assignment completion percentage
sleep_hours_per_day	Approximate sleep hours per day
internet_quality	Good / Average / Poor
parental_support	High / Medium / Low
final_score	Final score; regression target

4. Methodology

Data inspection: Review dataset dimensions, schema, summary statistics, missing values, and duplicate rows.

Data cleaning: Drop exact duplicate records. Fill numeric missing values with column medians and categorical missing values with the most frequent category. Clip percentage and score values to valid 0–100 ranges.

Exploratory analysis: Inspect score distribution, plot attendance against final score, plot study hours against final score, and calculate correlations for numeric indicators.

Attendance analysis: Group attendance into Low, Moderate, Good, and Excellent bands, then compare score counts, means, and medians.

Feature engineering: Create study bands and a pass flag using an illustrative threshold of final_score >= 40. Adjust this threshold to the real grading policy if using another dataset.

Modeling: Use an 80/20 train-test split. Apply median imputation and scaling to numeric features and most-frequent imputation plus one-hot encoding to categorical features within a scikit-learn pipeline.

Evaluation: Evaluate regression with MAE, RMSE, and R^2. Evaluate classification with accuracy and class-wise precision, recall, and F1-score.

5. Visualizations

Figure 1. Distribution of final scores.

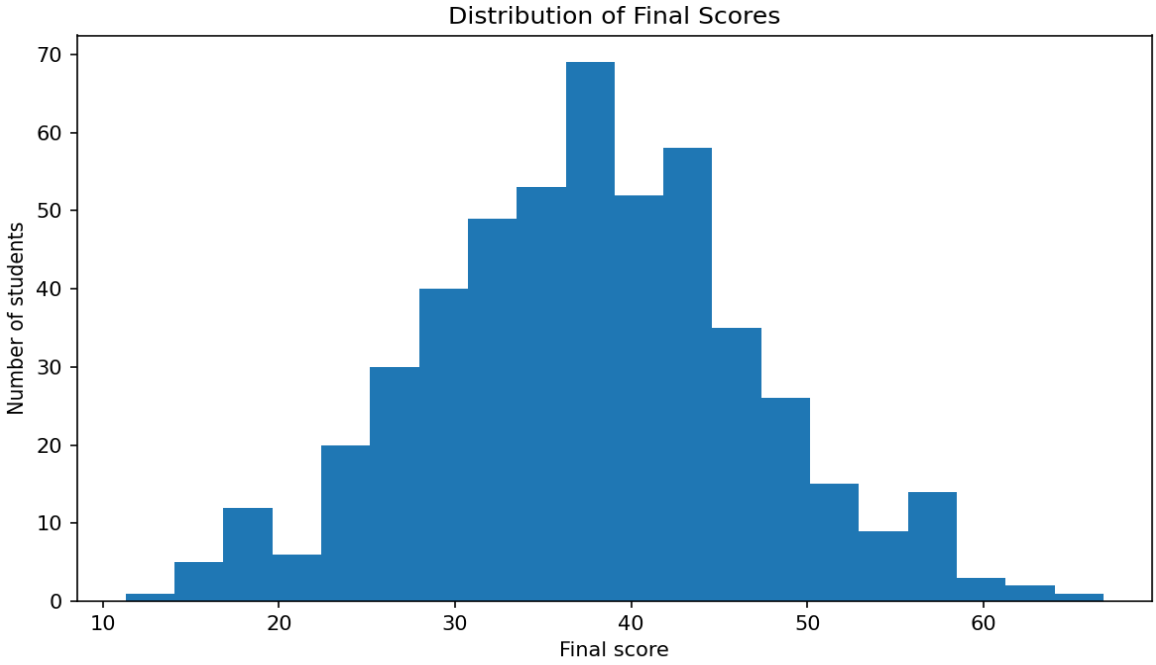


Figure 2. Attendance percentage versus final score.

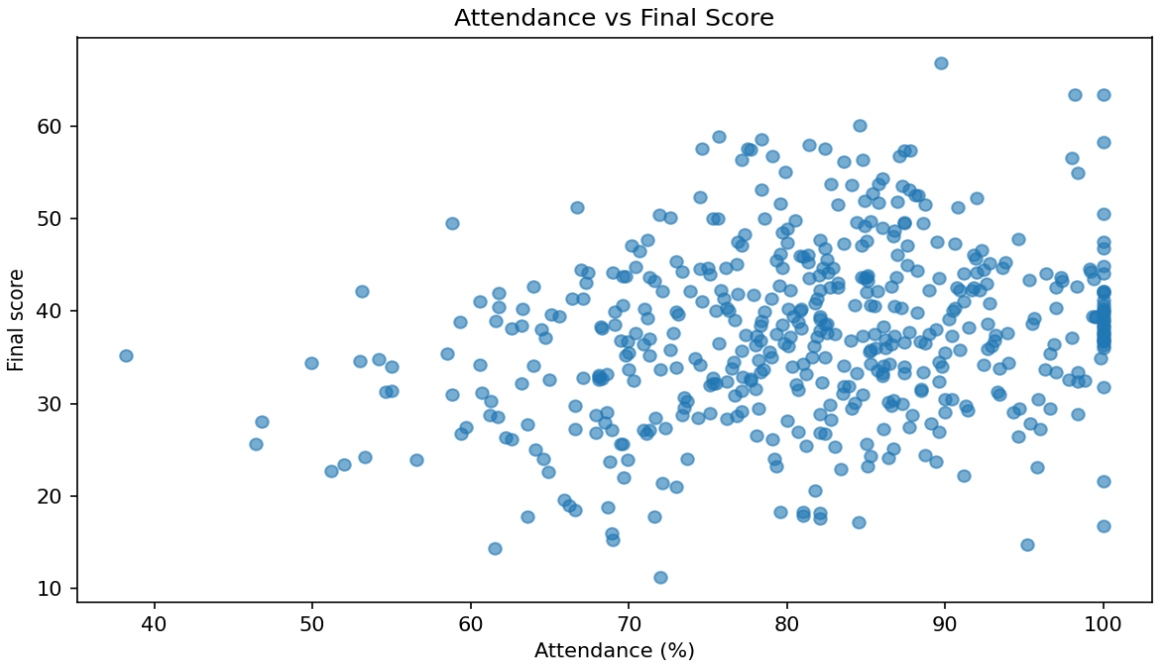


Figure 3. Study hours per day versus final score.

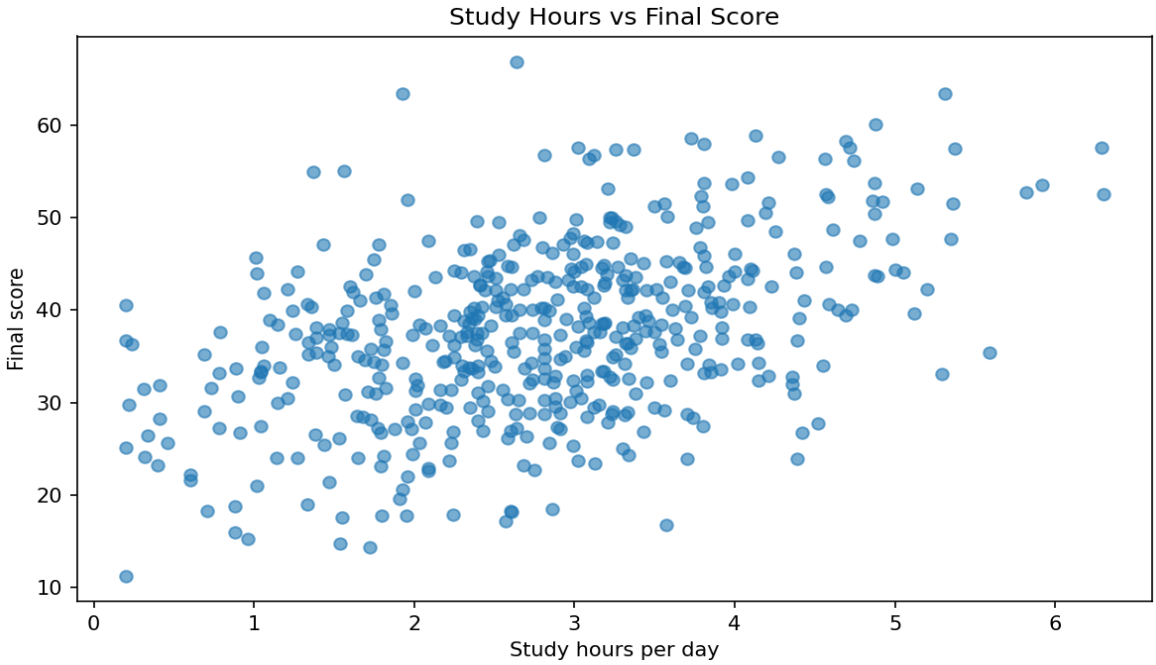
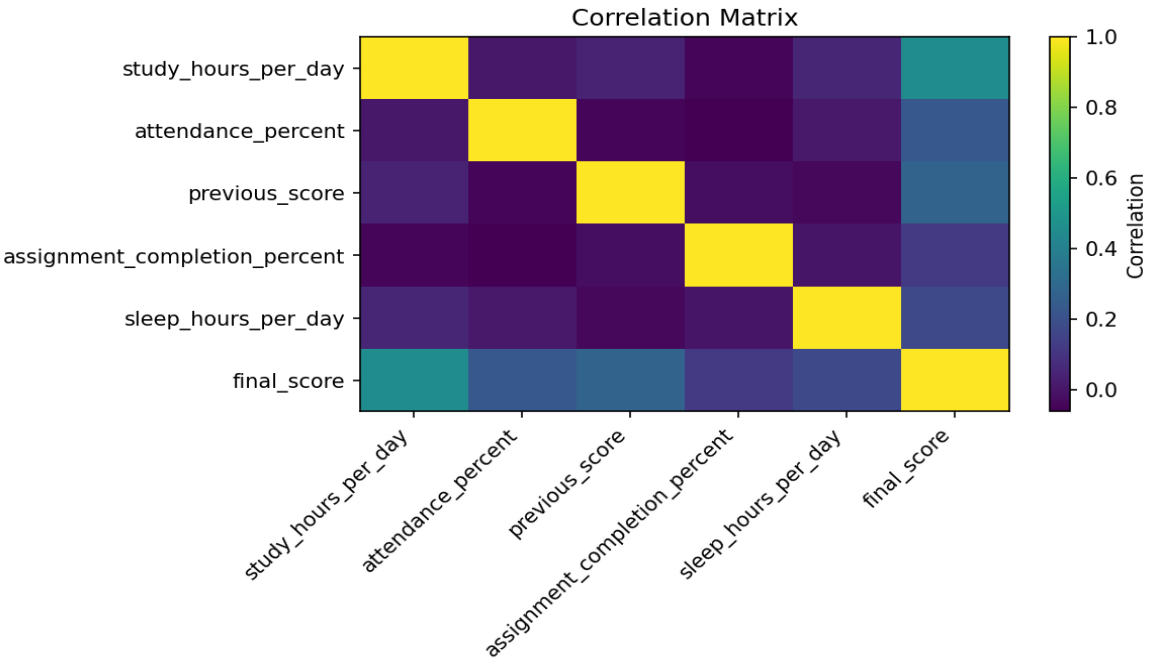


Figure 4. Correlation matrix for selected numeric variables.



6. Model Evaluation

Model	Metrics	Observed result on synthetic test split
Linear Regression	MAE	5.713
Linear Regression	RMSE	7.032
Linear Regression	R^2	0.407
Logistic Regression	Accuracy	0.710

Interpretation: MAE and RMSE express average prediction error in score points; lower is better. R^2 compares the model with a mean-prediction baseline. Classification accuracy is the share of correct pass/fail predictions. These results are only an illustration of the synthetic dataset and should not be generalized to real students.

7. Key Findings and Discussion

- The workflow demonstrates how to identify and address common data-quality issues before analysis.
- Attendance and study time can be explored alongside prior performance and assignment completion, but correlation does not prove causation.
- The regression and classification models provide baseline examples for a beginner portfolio; cross-validation and comparison against simple baselines would improve evaluation.
- The supplied brief did not contain subject-wise marks, so subject-wise strengths and weaknesses require additional valid subject columns.

8. Limitations and Ethical Considerations

- The dataset is synthetic; no claims should be made about actual schools, students, or learning outcomes.
- Do not use predictions as the sole basis for grading, admissions, discipline, or other high-impact decisions.
- For real data, obtain authorization, minimize personal information, secure files, and follow applicable privacy rules.
- Check errors and performance across relevant groups and involve educators when interpreting results.
- Document the data source, target definition, assumptions, and limitations.

9. Project Files

- data/student_performance_raw.csv — raw synthetic data with intentionally introduced missing values and duplicates.
- data/student_performance_clean.csv — cleaned dataset.
- data/student_performance_features.csv — cleaned dataset with attendance/study bands and pass label.
- notebooks/student_performance_learning_analytics.ipynb — annotated notebook covering the workflow.
- reports/ — plots, descriptive statistics, attendance summary, correlation matrix, and model evaluation.
- README.md and requirements.txt — setup, documentation, and dependencies.

10. How to Run

1. Extract the project ZIP. 2. Create and activate a Python virtual environment. 3. Install dependencies with `pip install -r requirements.txt`. 4. Launch Jupyter Notebook and open `notebooks/student_performance_learning_analytics.ipynb`. 5. Run cells from top to bottom.

Suggested GitHub Commit

Commit summary: Build student performance and learning analytics project

Description: Add synthetic dataset, data-cleaning workflow, EDA visualizations, feature engineering, regression and classification baselines, evaluation reports, and project documentation.